By manipulating the geometric series $\sum_{k=0}^{\infty} x^k = \frac{1}{1-x}$, $|x| < 1$, we were able to represent a restricted class of functions as power series. This representation was very useful, for instance, for integrating various functions whose antiderivatives cannot be expressed in terms of elementary functions.

Here we investigate more general questions:

(1) Which functions have power series representations?

(2) How can we find such representations?

We begin with the second question. Suppose f is a function which can be represented by a power series:

$$\begin{aligned} f(x) &= \sum_{k=0}^{\infty} c_k x^k, \quad |x-a| < R \\ &= c_0 + c_1(x-a) + c_2(x-a)^2 + c_3(x-a)^3 + \cdots + c_n(x-a)^n + \cdots, \quad |x-a| < R. \end{aligned}$$

Let's try to determine what the coefficients c_n must be in terms of f. To begin, notice that

$$\begin{aligned} f(a) &= c_0 + 0 + 0 + 0 + \ldots \\ &= c_0. \end{aligned}$$

Thus, $\boxed{c_0 = f(a)}$.

1.) Solve for c_1 by differentiating the power series and evaluating $f'(x)$ at $x = a$.

2.) Now consider $f''(a)$ to solve for c_2.

3.) Now consider $f^{(3)}(a)$ to solve for c_3. Do you see a pattern emerging? Write an expression for c_n.

We have proved the following:

Theorem. If f has a power series representation at a as

$$f(x) = \sum_{n=0}^{\infty} c_n (x-a)^n, \quad |x-a| < R,$$

then its coefficients are given by the formula

$$c_n = \frac{f^{(n)}(a)}{n!}.$$

(This formula also holds for $n = 0$ if we define $0! = 1$.) Thus, if f has a power series representation at a, then

$$\boxed{f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n}. \tag{1}$$

Definition. The series in (1) is called the *Taylor series of the function f centered at a*.

For the special case $a = 0$, the Taylor series becomes

$$\boxed{f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}x^n}, \tag{2}$$

which is called a *Maclaurin series*.

Remark. We have proved that *if* f can be represented as a power series about a, then f must be equal to the sum of its Taylor series. However, as we will see, there exist functions that are not equal to the sum of their Taylor series, and hence cannot be represented as a power series.

4.) Find the Maclaurin series of the function $f(x) = e^x$ and its radius of convergence.

We can therefore draw the conclusion that *if* e^x has a power series expansion centered at 0, then

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}, \quad x \in \mathbb{R}.$$

So how can we determine whether e^x *does* have a power series representation? We will answer this in the next lesson. Today, we will practice calculating Taylor series and finding the radius of convergence.

5.) Find the Maclaurin series for $\sin(x)$ and find its radius of convergence.

6.) Find the Maclaurin series for $\cos(x)$ and find its radius of convergence.